

System Architecture & Interface Specification (v3.1)

System Level High-Voltage, Networking, and Diagnostic Pinout Definitions

1. High-Voltage Interlock Loop (HVIL) Architecture

The HVIL subsystem monitors continuity of all high-voltage connectors and service disconnect covers.

- Section 3.4: HVIL signal generator is specified for 1 kHz PWM with 5 ms fault detector as described in architecture section 3.4.
- Diagnostic loop impedance is monitored via secondary microcontroller ADC input.

2. Inverter Field Oriented Control (FOC) Architecture

- Section 5.2 defines the FOC flux weakening vector control block architecture and current loop equations.
- Pulse-width modulation is synthesized via third-harmonic injection space vector generator.

3. High-Voltage Soft-Start Pre-Charge Circuit

- Section 4.3 defines the pre-charge resistor sizing (50 ohm) and soft-start timing diagram.
- Inrush current profile is limited during contactor sequencing.

4. Electric Vehicle Grid Interface Architecture

- Section 6.1 specifies the HomePlug GreenPHY architecture and TLS 1.3 cipher suite requirements.
- Smart grid interface aligns with ISO 15118-20 communications controller.

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5. Thermal Management Controller Interfaces

- Section 7.4 describes water pump LIN diagnostic messages and dry-run fault detection algorithm.
- Pump speed feedback is encoded as 8-bit LIN frame with cavitation alarm flag.

6. In-Vehicle Networking Socket Buffers

- Section 8.1 defines the static socket buffer allocation tables in AUTOSAR BswM configuration.
- Up to 8 concurrent TCP stream connections are allocated without dynamic memory allocation.

7. Diagnostic Fault Code Protocol Architecture

- Section 9.2 specifies negative response code handling tables for diagnostic stack.
- Handles NRC 0x12 (subFunctionNotSupported) and NRC 0x22 (conditionsNotCorrect).

8. Safety Clock Supervisor & Mechanical Shock Profiles

- Section 10.1 defines the clock supervisor hardware architecture and backup oscillator configuration.
- Section 11.3 describes the secure logging event format and flash partition layout.
- Section 12.1 specifies mechanical shock test parameters and fixture mounting details per ISO 16750-3.